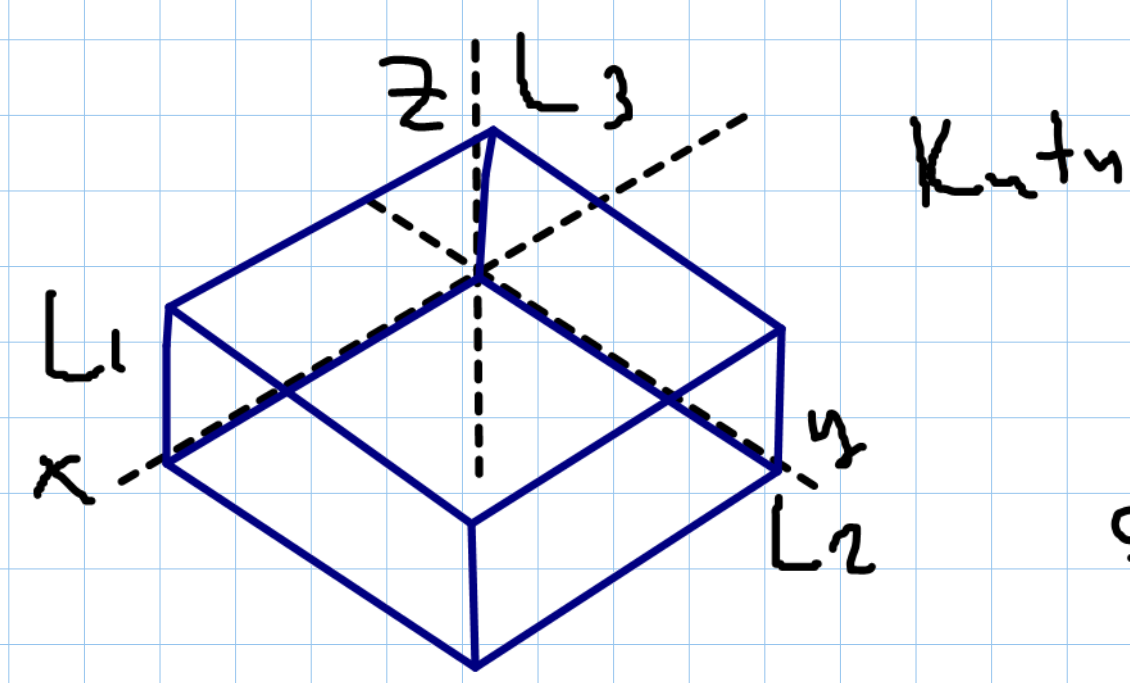


Kuantum Mekanik I Bölüm 9

Tekrar

Schr D'eq $H = \frac{p^2}{2m} + V(r)$

$HU = EU$



$V(x) = \infty$ for $x > L_1$ and $x < 0$

$V(x) = 0$ for $0 \leq x \leq L_1$

$V(y) = \infty$ for $y > L_2$ and $y < 0$

$V(y) = 0$ for $0 \leq y \leq L_2$

$V(z) = \infty$ for $z > L_3$ and $z < 0$

$V(z) = 0$ for $0 \leq z \leq L_3$

Bölüm 4'ten:

$U(x) = A \sin kx + B \cos kx$

$U(0) = 0 \Rightarrow A = 0$

$U_n(x) = \sqrt{\frac{2}{L}} \sin k_n x \Rightarrow U_n(x) = \sqrt{\frac{2}{L}} \sin k_n x$

$E_n = \frac{\hbar^2 k_n^2}{2mL^2} n^2$

$H = \frac{p_x^2}{2m} + \frac{p_y^2}{2m} + \frac{p_z^2}{2m} + V(x,y,z) \Rightarrow H = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} - \frac{\hbar^2}{2m} \frac{\partial^2}{\partial y^2} - \frac{\hbar^2}{2m} \frac{\partial^2}{\partial z^2} + V(x) + V(y) + V(z)$

$UE(x,y,z) = X(x)Y(y)Z(z)$

$\Rightarrow HUE = EU_E$

$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} (XYZ) - \frac{\hbar^2}{2m} \frac{\partial^2}{\partial y^2} (XYZ) - \frac{\hbar^2}{2m} \frac{\partial^2}{\partial z^2} (XYZ) + V(x)(XYZ) + V(y)(XYZ) + V(z)(XYZ) = E(XYZ)$

4 taraftan X, Y, Z

Böl

$-\frac{\hbar^2}{2m} X \frac{\partial^2 Y}{\partial y^2} - \frac{\hbar^2}{2m} Y \frac{\partial^2 X}{\partial x^2} - \frac{\hbar^2}{2m} Z \frac{\partial^2 X}{\partial z^2} + V(x)X + V(y)Y + V(z)Z = E_1 X + E_2 Y + E_3 Z$

$-\frac{\hbar^2}{2m} \frac{\partial^2 X}{\partial x^2} \left(\frac{1}{X}\right) + V(x) = E_1$

$X = \sqrt{\frac{2}{L_1}} \sin k_1 x$

$-\frac{\hbar^2}{2m} \frac{\partial^2 Y}{\partial y^2} \left(\frac{1}{Y}\right) + V(y) = E_2$

$Y = \sqrt{\frac{2}{L_2}} \sin k_2 y$

$-\frac{\hbar^2}{2m} \frac{\partial^2 Z}{\partial z^2} \left(\frac{1}{Z}\right) + V(z) = E_3$

$Z = \sqrt{\frac{2}{L_3}} \sin k_3 z$

$E_1 = \frac{\hbar^2 k_1^2}{2mL_1} n_1^2$ $E_2 = \frac{\hbar^2 k_2^2}{2mL_2} n_2^2$ $E_3 = \frac{\hbar^2 k_3^2}{2mL_3} n_3^2$

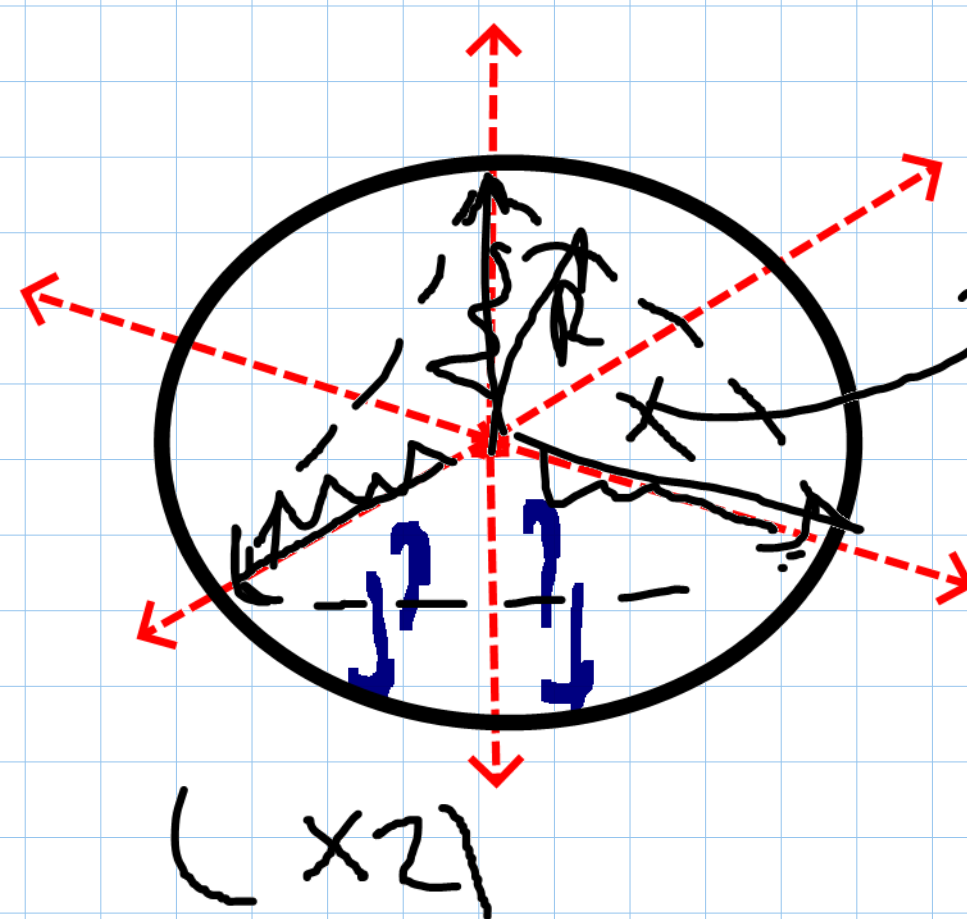
$U_{n_1 n_2 n_3}(x,y,z) = \sqrt{\frac{2}{L_1}} \sqrt{\frac{2}{L_2}} \sqrt{\frac{2}{L_3}} \sin k_1 x \sin k_2 y \sin k_3 z$

$E_{n_1 n_2 n_3} = \frac{\hbar^2 k^2}{2m} \left(\frac{n_1^2}{L_1^2} + \frac{n_2^2}{L_2^2} + \frac{n_3^2}{L_3^2} \right)$

$n = 1, 2, 3, \dots$

$E_{n_1 n_2 n_3}$	n_1	n_2	n_3	$U_{n_1 n_2 n_3}$
3A	1	1	1	$\left(\frac{2}{L}\right)^{3/2} \sin k_1 x \sin k_2 y \sin k_3 z$
6A	2	1	1	$\left(\frac{2}{L}\right)^{3/2} \sin 2k_1 x \sin k_2 y \sin k_3 z$
	1	2	1	$\left(\frac{2}{L}\right)^{3/2} \sin k_1 x \sin 2k_2 y \sin k_3 z$
	1	1	2	$\left(\frac{2}{L}\right)^{3/2} \sin k_1 x \sin k_2 y \sin 2k_3 z$
9A	2	2	1	
	2	1	2	
	1	2	2	
11A	3	1	1	
	1	3	1	
	1	1	3	

$n_1^2 + n_2^2 + n_3^2 = \frac{2mL^2}{\hbar^2} E \Rightarrow E = E_F$



$V = \frac{4\pi}{3} R^3$

=)

$n_1^2 + n_2^2 + n_3^2 = \frac{2mL^2}{\hbar^2} E_F$

$n_x^2 + n_y^2 + n_z^2 = R^2$ Küre Denk =)

$R^2 = \frac{2mL^2}{\hbar^2} E_F \Rightarrow R = \left(\frac{2mL^2}{\hbar^2} E_F \right)^{1/2}$

Kabuk V nin 8 de 1:

$V = \frac{4\pi}{3} R^3 \Rightarrow V = \frac{4\pi}{3} \left(\frac{2mL^2}{\hbar^2} E_F \right)^{3/2} \left(\frac{1}{8} \right) \times 2$

$$V = \frac{4}{3} \pi R^3 \Rightarrow V = \frac{4}{3} \pi \left(\frac{2mL^2}{\hbar^2 n^2} E_F \right)^{3/2} \left(\frac{1}{8} \right) (\times 2)$$

$$V = \frac{4}{3} \pi \left(\frac{2mL^2}{\hbar^2 n^2} E_F \right)^{3/2} \Rightarrow \frac{3V}{4\pi} \left(\frac{\hbar^2 n^2}{2mL^2} \right)^{3/2} = (E_F)^{3/2} \Rightarrow E_F = \frac{3V}{2\pi L^3} \left(\frac{\hbar^2 n^2}{2m} \right)^{3/2} = E_F$$

$$\Rightarrow E_F = \left(\frac{3V}{4\pi L^3} \right)^{2/3} \left(\frac{\hbar^2 n^2}{2m} \right)$$

$$E_F = \left(\frac{3n}{4\pi} \right)^{2/3} \frac{\hbar^2 n^2}{2m} \quad n = \frac{V}{L^3} \left(\frac{\hbar^2 m}{4\pi} \right)$$

$$\Rightarrow \frac{1}{8} \int d^3 n = \frac{\hbar^2 n^2}{mL^2} \left(\frac{1}{8} \right) \int n^2 d^3 n = \frac{\hbar^2 n^2}{8mL^2} 4\pi \int_0^R n^4 dn = \frac{\hbar^2 n^2}{10mL^2} R^5$$

